

Exposing Generalization Gaps in Industrial Anomaly Detection under Distribution Shift via Maximum Discrepancy Competition

Jiajin Xu , Haifeng Liao , Boyang Wan ^(✉) , Xiujuan Zhao ^(✉) , and Fan Yang ^(✉) 

Jiangxi Science and Technology Normal University, Nanchang 330038, China
wanboyangjerry@jxstnu.edu.cn, xjzhao@jxstnu.edu.cn, fyang@jxstnu.edu.cn

Abstract. Industrial anomaly detection (IAD) has advanced rapidly, yet evaluation still relies on fixed benchmarks drawn from the same distribution as the training data. These benchmarks are approaching saturation and more fundamentally reveal little about generalization under distribution shift. While some benchmarks do target distribution shift, they rely on fixed, pre-annotated data and cannot adaptively select the most discriminating samples. To address this issue, we introduce the Distribution-Shift Maximum Discrepancy (DS-MAD) Competition. Specifically, from a large-scale pool of unlabeled images captured under shifted conditions, DS-MAD adaptively selects contentious samples, those on which model pairs disagree most, and annotates only this compact subset to compute pairwise I-AUROC scores. These pairwise comparisons are then aggregated via the Perron rank method into a dual ranking that captures both absolute generalization and relative robustness in direct model-to-model comparison. Our analysis yields several findings. Reverse knowledge distillation methods are most robust under distribution shift, while top-ranked models in normalizing flow, diffusion, and one-class paradigms suffer the sharpest degradation, suggesting that benchmark overfitting rather than architectural choice drives poor generalization. The dual ranking further reveals systematic divergences largely invisible to single-metric evaluation. We release the code and datasets at <https://github.com/hbqdcq-hx/DS-MAD>.

Keywords: Industrial Anomaly Detection · Maximum Discrepancy Competition · Distribution Shift · Generalization Evaluation · Model Comparison

1 Introduction

Industrial anomaly detection (IAD) is central to quality inspection across many industries [12]. Anomalous samples are rare and expensive to collect, so most existing IAD methods train exclusively on normal images. They flag any departure from the learned normality as anomalous. Off-the-shelf IAD benchmarks such as MVTec AD [4], BTAD [18], and VisA [33] support the development of

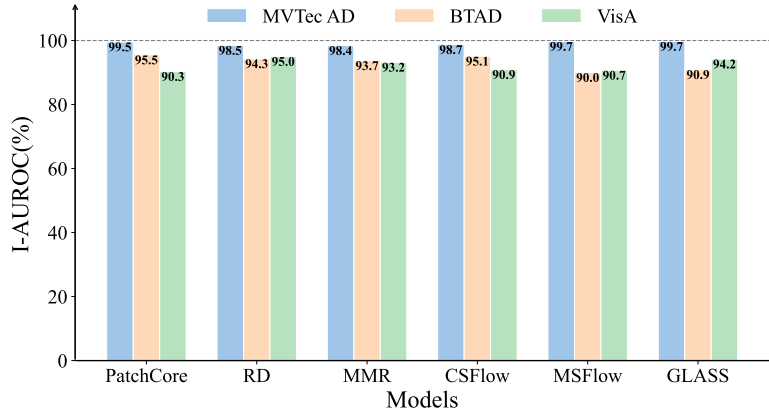


Fig. 1. Image-level I-AUROC of representative IAD methods on three popular benchmarks.

numerous methods [13,20,6]. As shown in Fig. 1, representative IAD methods now achieve near-saturated image-level I-AUROC across all three benchmarks, reinforcing the impression that anomaly detection has been largely solved.

However, such benchmark performance reveals little about reliability under production conditions. MVtec AD was acquired under highly controlled laboratory settings with limited variation in defects, lighting, and backgrounds. Even benchmarks collected from real production lines, such as BTAD, represent a fixed set of conditions that cannot capture the full diversity of deployment environments. Production environments, by contrast, are subject to persistent distribution shift. This gap between benchmark conditions and deployment environments defines distribution shift. Whether existing models generalize under such shift remains an open question [5,7].

Standard evaluation protocols with fixed benchmarks share the same distribution as the training data and reveal little about generalization under shift. Building a labeled real-world test set demands exhaustive annotation, which is often infeasible in industrial settings [2]. Random sampling selects easy images that fail to discriminate between state-of-the-art models. Existing work on distribution shift in IAD faces similar limitations. GNL [5] improves model robustness to distribution shift but does not provide a framework for systematic model comparison. Beyond Benchmarks [2] reveals rank reversals between benchmark and real-world performance, yet relies on fixed multi-dataset evaluation. AnoShift [7] offers a temporal shift benchmark protocol but targets network intrusion detection rather than industrial vision. All three share a common weakness: they depend on fixed or pre-annotated data and cannot adaptively select the most discriminating samples.

The MAXimum Discrepancy (MAD) Competition directly addresses this weakness. It adaptively selects, from a large pool of unlabeled images, the samples

on which two models most strongly disagree. These contentious samples carry the most information for distinguishing the models [27,17]. MAD has successfully exposed hidden weaknesses in image classification [26], semantic segmentation [29], image captioning [25], and large language model evaluation [8]. It has never been applied to industrial anomaly detection. It has never been used under distribution shift. To address this, we introduce the **Distribution-Shift MAXimum Discrepancy (DS-MAD) Competition**, the first framework to combine MAD with distribution-shift evaluation for systematic comparison of IAD model generalization. We start from a pool of 43069 unlabeled, open-domain images and use DS-MAD to select those on which two models produce the most divergent anomaly scores. We collect human labels only for these contentious samples. We then compute pairwise I-AUROC scores and aggregate them into a global ranking via the Perron rank method [21]. The framework is incremental: adding a new model or new data does not require re-running the full pipeline. We apply DS-MAD to 14 representative models across 15 industrial categories and find that every model loses over 30% I-AUROC under distribution shift. Benchmark rankings from MVTec AD diverge systematically from the generalization rankings produced by DS-MAD on real-world data. The main contributions of this work are threefold: (1) **The DS-MAD Framework**. We introduce DS-MAD, the first framework to adapt the MAD Competition to industrial anomaly detection. It combines adaptive sample selection with distribution-shift evaluation, to systematically compare model generalization without exhaustive dataset labeling. (2) **A Distribution-Shift Stress Test Set**. We construct and annotate a contentious test set $\mathcal{D}$ from 43069 real-world images spanning 15 product categories. These images cover the same categories as MVTec AD but exhibit substantially greater variation in environment and appearance. (3) **Empirical Evidence of Generalization Gaps**. We provide the first large-scale evaluation of 14 state-of-the-art IAD models under distribution shift. All models lose over 30% I-AUROC, and benchmark rankings diverge systematically from generalization rankings.

2 Related Work

2.1 Unsupervised Industrial Anomaly Detection

Unsupervised IAD methods are broadly divided into feature embedding and image reconstruction approaches [13]. Feature embedding methods model the distribution of normal samples in a feature space extracted by a pretrained backbone. Anomalies are detected as deviations from this nominal distribution. Four paradigms dominate. Memory bank methods, exemplified by Patch-Core [20], construct a reference bank of nominal patch features using coresets sampling. Teacher-student architectures train a student network to match a pretrained teacher on normal images, producing larger discrepancies on anomalies. In the reverse knowledge distillation variant (RD [6], RD++ [24], UniNet [28], URD [14]), the student acts as a decoder that reconstructs the teacher’s encoded features from a compact bottleneck: normal features are reconstructed well, while

anomalous features produce large errors. DeSTSeg [31] uses a self-distillation mechanism. Normalizing flow models such as CFLOW-AD [9] and MSFlow [32] map high-dimensional features to a standard Gaussian distribution via invertible transformations, directly estimating data likelihood. One-class classification methods learn a decision boundary between normal and anomalous features. SimpleNet [15] trains a discriminator on synthetic anomalies, and CoMet-SN [1] incorporates meta-learning to handle ambiguous samples near the boundary. Reconstruction methods learn a mapping from normal images to their reconstructions. Anomalous regions produce larger reconstruction errors. This category spans autoencoders, GANs, Transformers, and diffusion models [13]. DSR [30] uses a dual-decoder architecture with quantized feature representations. Dinomaly [10] adopts a pure Transformer architecture relying solely on attention and MLPs. For diffusion models, DeCo-Diff [3] improves the denoising process through deviation correction, and InvAD [22] bypasses conventional denoising by directly inferring latent variables via DDIM inversion. GAN-based architectures have rarely been adopted in recent unsupervised IAD research and are not included in our evaluation [13].

2.2 Maximum Discrepancy Competition for Model Evaluation

Traditional model evaluation relies on fixed datasets that offer sparse coverage of high-dimensional spaces [27]. The MAXimum Discrepancy (MAD) Competition addresses this by adaptively selecting contentious samples from a large pool of unlabeled data [27]. gMAD [17] extended MAD to multi-model comparison with statistical aggregation and adaptive sampling, achieving stable rankings with roughly 15 sample pairs. Wang et al. [26] replaced gradient synthesis with Top-K sampling from a data pool, making MAD applicable to arbitrary black-box models, and extended it to image classification. MAD has since been further extended to semantic segmentation [29], image captioning [25], and large language model evaluation [8], each time exposing weaknesses invisible to fixed benchmarks. Most recently, PMDC [16] explored using LLM oracles to replace human annotation. Critically, all prior MAD work operates within the same data distribution as the models being tested. None has evaluated model generalization under distribution shift.

2.3 Anomaly Detection under Distribution Shift

Distribution shift poses a challenge for anomaly detection: the discrepancy between training and test distributions causes sharp performance degradation. Existing work follows two main directions: model improvement and benchmark construction. On the model improvement side, GNL [5] introduces distribution-invariant normality learning on top of the RD4AD architecture, improving OOD I-AUROC by over 10%. These methods, however, aim to improve individual models and do not provide a framework for systematic comparison of generalization across models. On the benchmark side, AnoShift [7] proposes a temporal shift benchmark over the Kyoto-2006+ network traffic dataset, introducing IID,

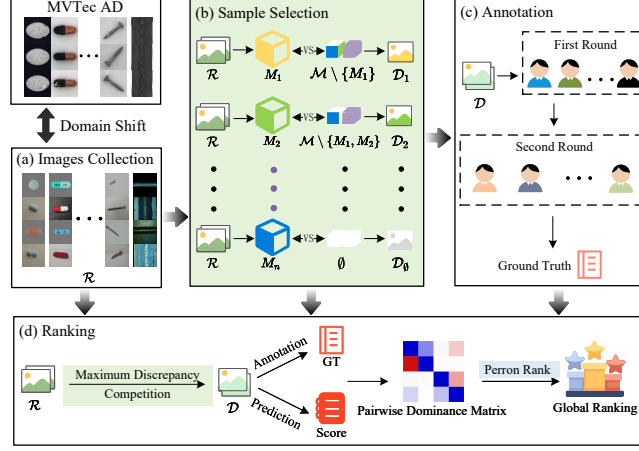


Fig. 2. Overview of the proposed framework. Our approach consists of four sequential modules: (a) Raw Test Set Collection; (b) Contentious Sample Selection; (c) Anomaly Detection Annotation Experiment; (d) Ranking based Generalization Evaluation.

NEAR, and FAR splits to quantify performance degradation over time. Beyond Benchmarks [2] offers a systematic critique of MVTEC AD, revealing rank reversals on real production data and center-crop bias. These benchmarks use fixed datasets or predefined temporal splits and cannot adaptively select the most discriminating samples. All rely on pre-annotated or fixed-split data. Combining adaptive sample selection with distribution-shift evaluation to systematically compare IAD model generalization remains an open direction.

3 Proposed Method

DS-MAD operates the MAD Competition under distribution shift. Unlike prior MAD methods that select contentious samples from the same distribution as the training data, DS-MAD constructs a raw image pool $\mathcal{R}$ from heterogeneous, open-domain sources, introducing distribution shift into the evaluation pipeline by design. The framework proceeds in four stages, as illustrated in Fig. 2: (1) collecting $\mathcal{R}$ from diverse sources; (2) applying pairwise MAD competition to select the most contentious samples from $\mathcal{R}$; (3) conducting a two-round expert annotation experiment on the selected samples; and (4) aggregating pairwise comparisons into a global generalization ranking via the Perron rank method.

3.1 Raw Test Set Collection

To introduce distribution shift, we construct $\mathcal{R}$ to differ from MVTEC AD across three dimensions: acquisition hardware, imaging conditions, and defect diversity. MVTEC AD was acquired using a single industrial sensor under controlled

illumination, with defects created manually in a laboratory setting. By contrast, the Roboflow database [19] provides crowd-sourced images captured by diverse consumer cameras under uncontrolled lighting and backgrounds, introducing hardware and environmental variation absent from MVTec AD. We collect images covering bottle, cable, capsule, screw, grid, hazelnut, toothbrush, transistor, metal nut, pill, wood, and zipper. Tianchi competition data¹² come from real factory production lines, where defects occur naturally rather than being manually created, further diversifying defect appearance. We use these for the tile and carpet categories. For leather and tile, we supplement with self-captured images using a different industrial camera and lens, captured under conditions that differ from the MVTec AD setup to reflect cross-factory variation.

After collection, we filter out images with a length or width below 512 pixels (the maximum input size among the candidate models) and remove duplicates. We also conduct manual filtering to discard deformed images. This process produces the raw set $\mathcal{R}$ containing 43069 images. Together, the three heterogeneous sources ensure that $\mathcal{R}$ spans the variability of real-world deployment that a single-source laboratory benchmark cannot capture.

3.2 Contentious Sample Selection

To construct a discriminative test set $\mathcal{D}$ from $\mathcal{R}$, we employ a pairwise MAD strategy. We begin with the simplest case of two models M_a and M_b . Given the raw pool $\mathcal{R}$, which exhibits substantial distribution shift relative to MVTec AD, we seek the images on which M_a and M_b disagree most. These contention points carry the most information for comparing model generalization under shift. For each pair of models (M_a, M_b) , we compute a discrepancy score for every image $\mathbf{x}_i \in \mathcal{R}^c$ (where $\mathcal{R}^c$ denotes the subset of $\mathcal{R}$ for category c) as the absolute difference of their anomaly scores. The Top- k images with the highest discrepancy scores within each category are then selected and aggregated across all categories to form the contentious set $\mathcal{D}_{a,b}$:

$$\mathcal{D}_{a,b} = \bigcup_{c=1}^{15} \left(\text{Top-}k \mid M_a(\mathbf{x}_i) - M_b(\mathbf{x}_i) \mid \right)_{\mathbf{x}_i \in \mathcal{R}^c} \quad (1)$$

where $M_a(\mathbf{x}_i)$ and $M_b(\mathbf{x}_i)$ denote the anomaly scores of model M_a and M_b on image $\mathbf{x}_i$. The final contentious test set $\mathcal{D}$ is the union of all pairwise subsets: $\mathcal{D} = \bigcup_{a \neq b} \mathcal{D}_{a,b}$. Given J models, this process generates up to $\frac{15 \cdot k \cdot J(J-1)}{2}$ candidate samples, and the human annotation cost depends only on k and J rather than on $\mathcal{R}$. The discrepancy scores, computed on distribution-shifted samples from $\mathcal{R}$, capture both model disagreement and differential sensitivity to distribution change. Since different model pairs may select overlapping samples, the actual

¹ Smart diagnosis of cloth flaw dataset (2020). <https://tianchi.aliyun.com/dataset/dataDetail?dataId=79336>

² Ceramic tile defect detection dataset (2021). <https://tianchi.aliyun.com/dataset/110088>

size of $\mathcal{D}$ can be smaller than the theoretical upper bound. $\mathcal{R}$ can therefore be expanded arbitrarily without increasing annotation costs to cover a wider range of distribution shifts.

3.3 Anomaly Detection Annotation Experiment

To obtain reliable ground-truth labels for human-aligned evaluation, we conduct a two-round subjective annotation experiment on the contentious set $\mathcal{D}$. First, we provide unified annotation guidelines that clearly define normal and anomalous samples, and all annotators receive sufficient training to ensure a consistent understanding of industrial defect characteristics before formal annotation. In the first round, nine annotators with backgrounds in computer vision and industrial inspection independently label each sample; an image is assigned a final label only if at least seven annotators produce identical results, otherwise it proceeds to the second round. In the second round, six annotators discuss and reach a consensus to determine the final label. Only 1.27% of the samples required this second round, and the first-round inter-annotator agreement exceeded 77.8% (7 out of 9). The two-round protocol is particularly important under distribution shift. Images drawn from such heterogeneous sources often contain ambiguous cases, subtle defects in unfamiliar lighting, or normal variation that resembles anomalies, challenging even trained annotators. The consensus mechanism ensures that the ground-truth labels for these distribution-shifted samples are reliable.

3.4 Ranking based Generalization Evaluation

Pairwise comparison reveals maximum discrepancies between model pairs but does not directly yield a global ranking, as intransitive phenomena may occur. Intransitivity is exacerbated under distribution shift: a model that performs well on in-distribution data may fail on shifted samples where another model excels, creating cyclic dominance patterns that simple averaging cannot resolve. The Perron rank method, by computing the principal eigenvector of the pairwise dominance matrix, provides a principled way to aggregate such intransitive comparisons into a consistent global ranking. We adopt the Perron rank method [21] and aggregate the results through a three-step procedure. First, we construct the pairwise performance matrix $\mathbf{P}$. Its entries $p_{a,b}$ and $p_{b,a}$ are the I-AUROC of M_a and M_b , respectively, on their shared contentious subset $\mathcal{D}_{a,b}$ (see Appendix Fig. S1). Next, we derive the pairwise dominance matrix $\mathbf{F}$. An entry $f_{a,b} > 1$ indicates that M_a outperforms M_b on their contentious samples, while $f_{a,b} < 1$ indicates the opposite:

$$f_{a,b} = \frac{p_{a,b}}{p_{b,a}} \quad (2)$$

Finally, we compute the principal eigenvector (Perron vector) $\mathbf{q}$ of $\mathbf{F}$ via the power iteration method. The resulting vector $\mathbf{q} = [q_1, \dots, q_J]^\top$ gives the distribution-

Table 1. Comparison of per-category image counts between the MVTec AD test set and $\mathcal{R}$.

Category	MVTec AD	$\mathcal{R}$	Ratio
Screw	480	1344	2.8
Leather	369	1250	3.4
Bottle	292	1436	4.9
Transistor	313	1530	4.9
Tile	347	1813	5.2
Wood	326	1690	5.2
Hazelnut	501	2985	6.0
Grid	342	2225	6.5
Toothbrush	102	724	7.1
Cable	374	3203	8.6
Carpet	397	3654	9.2
Capsule	351	3770	10.7
Pill	434	5487	12.6
Metal nut	335	4548	13.6
Zipper	391	7410	19.0
Average	357	2871	8.04

shift generalization ranking, where a larger q_i corresponds to better generalizability for M_i under shift:

$$\mathbf{q}^{(t+1)} = \frac{\mathbf{F}\mathbf{q}^{(t)}}{\|\mathbf{F}\mathbf{q}^{(t)}\|_1}, \quad t = 0, 1, 2, \dots \quad (3)$$

Starting from any positive initial vector, the iteration converges to the Perron vector, which gives the global generalization ranking.

4 Experiments

4.1 Experimental Setup

Construction of the Raw Set $\mathcal{R}$ We construct $\mathcal{R}$ as described in Sec. 3.1, collecting 724 to 7410 images per category (43069 total, roughly eight times the size of MVTec AD). Table 1 gives the per-category breakdown.

Evaluated Unsupervised Industrial Anomaly Detection models We evaluate 14 IAD methods from 2022 to 2026: PatchCore [20] (memory bank); DeSTSeg [31], RD [6], RD++ [24], URD [14], and UniNet [28] (teacher-student); CFLOW-AD [9] and MSFlow [32] (normalizing flow); SimpleNet [15] and CoMet-SN [1] (one-class); DSR [30] (autoencoder); Dinomaly [10] (Transformer); and DeCo-Diff [3], InvAD [22] (diffusion). GAN-based methods are rarely used in

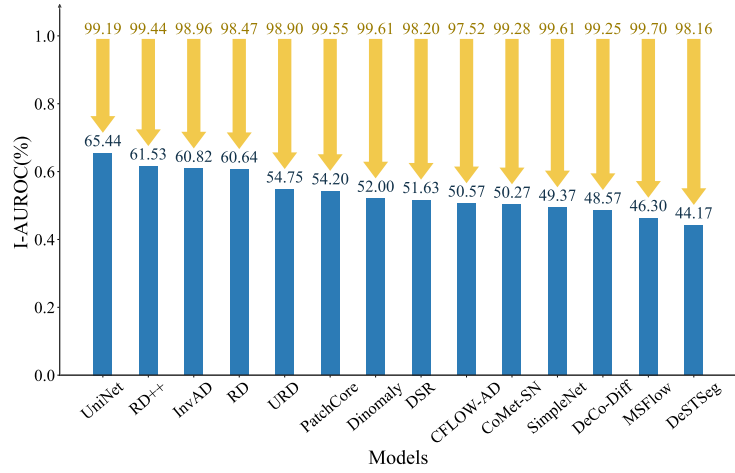


Fig. 3. Comparison of I-AUROC on MVTec AD (yellow) and $\mathcal{D}$ (blue).

IAD [13] and are excluded. All models use their official implementations; reproduced MVTec AD I-AUROC values fall within one percentage point of the published numbers. Anomaly score distributions differ widely across architectures, so we apply per-category min-max normalization before computing discrepancy scores and pairwise dominance ratios.

Subjective Annotation Experiment We run the MAD competition with $k = 20$ (Sec. 3.2), producing a set of contentious samples. After deduplication, $\mathcal{D}$ contains 8784 images, annotated via the two-round protocol of Sec. 3.3. The distribution of normal and anomalous samples is in Appendix Table S1.

4.2 Experimental Results

Comparison of gap between MVTec AD test set and $\mathcal{D}$ Fig. 3 compares I-AUROC of all 14 methods on the contentious test set $\mathcal{D}$ and the standard MVTec AD benchmark. All models lose substantial I-AUROC on $\mathcal{D}$, with declines exceeding 30% and reaching 53.40 percentage points for MSFlow, from 99.70% on MVTec AD to 46.30% on $\mathcal{D}$. UniNet, RD++, InvAD, URD and RD achieve the highest absolute I-AUROC on $\mathcal{D}$. These declines produce a systematic reshuffling of model rankings, as shown in Table 2.

Ranking results Table 2 reports the I-AUROC ranking of each model on MVTec AD ($\text{Rank}^{\clubsuit}$), on $\mathcal{D}$ ($\text{Rank}^{\star}$), and in the Perron global ranking ($\text{Rank}^{\dagger}$). The three rankings differ substantially. The top MVTec AD model (MSFlow) falls to 13th on both $\mathcal{D}$ and the Perron ranking ($\Delta_1 = \Delta_2 = -12$). UniNet moves from 7th to 1st on both ($\Delta_1 = \Delta_2 = +6$). The Spearman correlation

Table 2. Rank[♣] denotes the MVTec AD ranking, Rank[★] the $\mathcal{D}$ ranking, and Rank[†] the Perron global ranking. $\Delta_1 = \text{Rank}^{\clubsuit} - \text{Rank}^{\star}$ and $\Delta_2 = \text{Rank}^{\clubsuit} - \text{Rank}^{\dagger}$. Ties are assigned the same rank (dense ranking).

Models	Rank [♣]	Rank [★]	Δ_1 Rank	Rank [†]	Δ_2 Rank
MSFlow	1	13	-12	13	-12
Dinomaly	2	7	-5	9	-7
SimpleNet	2	11	-9	10	-8
PatchCore	3	6	-3	8	-5
RD++	4	2	+2	2	+2
CoMet-SN	5	10	-5	6	-1
DeCo-Diff	6	12	-6	5	+1
UniNet	7	1	+6	1	+6
InvAD	8	3	+5	3	+5
URD	9	5	+4	12	-3
RD	10	4	+6	4	+6
DSR	11	8	+3	7	+4
DeSTSeg	12	14	-2	14	-2
CFLOW-AD	13	9	+4	11	+2

between the MVTec AD ranking and either the $\mathcal{D}$ ranking or the Perron ranking is near zero. Benchmark performance has no meaningful relationship with generalization under distribution shift. Δ_1 measures the shift from MVTec AD to $\mathcal{D}$, Δ_2 the shift to the Perron ranking. The two metrics are complementary: Δ_1 reflects aggregate I-AUROC on the full contentious set, while Δ_2 isolates the most discriminating samples for each model pair.

4.3 Discussion

Architectural Analysis of Generalization under Distribution Shift First, **teacher-student architectures, and reverse knowledge distillation in particular, consistently improve their ranking under distribution shift.** UniNet (+6 on both Δ_1 and Δ_2), RD (+6/+6), RD++ (+2/+2), and URD (+4 on Δ_1), the four reverse knowledge distillation methods, all gain positions on $\mathcal{D}$ and occupy four of the top five spots. Reverse knowledge distillation has the student decoder reconstruct the encoded features of the teacher from a compact bottleneck: normality representations are anchored to semantic features rather than low-level image statistics that shift across domains. All four reverse KD methods rank in the top five on $\mathcal{D}$. Notably, URD gains 4 positions on Δ_1 but drops 3 on Δ_2 , exposing pairwise weaknesses invisible to aggregate metrics, a pattern that may stem from its multi-teacher ensemble design. DeSTSeg, the only teacher-student method that uses self-distillation rather than reverse KD, anchors the bottom of both rankings ($\Delta_1 = \Delta_2 = -2$). The consistent dominance of reverse KD implementations suggests that this paradigm is inherently robust to distribution shift.

Second, a cluster of high-ranking models across normalizing flow, diffusion, and one-class paradigms suffers the sharpest degradation under distribution shift. MSFlow (flow, #1 on MVTec AD, 99.70%) drops to 13th on $\mathcal{D}$ ($\Delta_1 = \Delta_2 = -12$), losing 53.40 percentage points of I-AUROC. SimpleNet (one-class, #2, 99.61%) falls to 11th ($\Delta_1 = -9$, $\Delta_2 = -8$). DeCo-Diff (diffusion, #6) declines to 12th ($\Delta_1 = -6$). Across three paradigms, these models share the same failure mode: tightly fitted normality boundaries on MVTec AD that cannot accommodate the variation in $\mathcal{D}$. Three methods within these paradigms resist this pattern. CFLOW-AD (also normalizing flow) gains 4 positions on Δ_1 and 2 on Δ_2 , starting from a low MVTec AD rank of 13th. Its weaker benchmark fit (97.52% vs. 99.70% for MSFlow) leaves more room for shifted normal samples. CoMet-SN (also one-class) drops only 5 positions on Δ_1 and one on Δ_2 , compared to -9 and -8 for SimpleNet. The meta-learning in CoMet-SN exposes it to distribution variation during training; the synthetic anomalies in SimpleNet overfit to benchmark-specific patterns. InvAD (also a diffusion model) achieves positive gains on both metrics ($\Delta_1 = \Delta_2 = +5$), unlike the steep decline of DeCo-Diff ($\Delta_1 = -6$); its DDIM inversion mechanism may be more robust to distribution shift than conventional denoising. The within-paradigm contrasts point to a common conclusion: features that maximize benchmark I-AUROC often fail under distribution shift, while methods incorporating robustness mechanisms, whether looser normality fitting, meta-learning, or DDIM inversion, trade benchmark points for real-world tolerance.

Third, the divergence between Δ_1 and Δ_2 shows that single-metric evaluation misses important structure in model generalization. DeCo-Diff ($\Delta_1 = -6$, $\Delta_2 = +1$) and URD ($\Delta_1 = +4$, $\Delta_2 = -3$) shift by 7 positions depending on the metric. Δ_1 aggregates I-AUROC across the full contentious set $\mathcal{D}$, pooled from all model pairs. Δ_2 derives from the Perron eigenvector of the pairwise dominance matrix $\mathbf{F}$ (Appendix Fig. S2), where each $f_{a,b}$ uses only $\mathcal{D}_{a,b}$, the samples that best separate M_a from M_b . This eigenvector-based synthesis, like the Kendall-Wei ranking method [11], recursively incorporates opponent strength into the final ranking. A model can score well on broadly shared samples yet struggle on its own most discriminating comparisons. Conversely, models where Δ_1 and Δ_2 agree, such as RD (+6/+6), UniNet (+6/+6), InvAD (+5/+5), and MSFlow ($-12/-12$), behave consistently regardless of metric. These complementary signals show that a full picture of generalization requires both absolute and relative robustness. Simi-

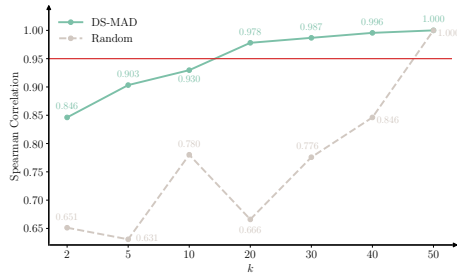


Fig. 4. Spearman correlation of rankings under different k settings within DS-MAD and with random sampling.

ilarly, models where Δ_1 and Δ_2 disagree, such as DeCo-Diff ($-6/+1$) and URD ($+4/-3$), show inconsistent performance across metrics. These complementary signals show that a full picture of generalization requires both absolute and relative robustness. Simi-

lar discrepancies between fixed-benchmark rankings and pairwise rankings have been documented in LLM evaluation [23], suggesting that this is a general phenomenon not specific to IAD.

Ranking Stability and Efficiency To evaluate ranking stability, we varied k from 2 to 50, using the ranking at $k = 50$ as the reference. As shown in Fig. 4, the DS-MAD ranking stabilizes rapidly: the correlation increases with k , surpassing 0.95 at $k = 20$ and reaching 0.978, after which growth slows. In contrast, random sampling produces low and unstable correlations across most k values, with no coefficient exceeding 0.90 except at $k = 50$. Random sampling fails to capture meaningful points of model disagreement, producing noisy comparisons that cannot reliably discriminate between models. Beyond ranking fidelity, DS-MAD achieves substantial annotation reduction. With $k = 20$, only 8784 unique images are selected for annotation from a pool of 43069, a cost reduction of approximately 79.6%. The annotation cost depends only on k and the number of models J , not on the size of $\mathcal{R}$, so the raw pool can be expanded to cover a wider range of distribution shifts without increasing human labeling effort.

Case Study: Background Interference The preceding analysis attributes rank changes to architectural factors. We now examine a complementary question from the data perspective: what specific form of distribution shift drives the observed degradation? The cable category is the primary source of low I-AUROC across all models. Observation of the high-discrepancy samples selected by MAD indicates that background interference is the dominant factor behind this degradation. To test this, we segmented cable images and replaced their original backgrounds with a uniform black background. The relative improvement in average I-AUROC on the cable category was 82.69%, confirming that background variation, a common form of distribution shift in real-world deployment, is a substantial source of performance degradation.

Limitations DS-MAD has several limitations. First, like all MAD-based methods, it provides relative rankings rather than absolute performance measures, and should complement, not replace, fixed-benchmark evaluation. Second, the pairwise selection requires $O(J^2)$ comparisons, which can become costly as the number of models grows, though the annotation cost remains bounded by k and J . Third, the annotation protocol assumes that trained annotators can reliably distinguish distribution-shifted anomalies, yet the background interference experiment suggests that even simple distribution changes may challenge model behavior; whether annotators are similarly affected warrants further study. Fourth, our evaluation is limited to industrial object categories derived from MVTec AD, and generalization to other domains remains to be validated.

5 Conclusion

In this paper, we have introduced DS-MAD, the first framework that operates the MAD Competition under distribution shift to evaluate IAD model generalization. We find that high I-AUROC on MVTec AD does not predict how well a model generalizes: the correlation between benchmark rankings and generalization rankings is weak. DS-MAD achieves stable rankings at modest annotation cost by adaptively selecting contentious samples. Its dual ranking captures both absolute performance decline and pairwise competitiveness. Three cross-paradigm findings emerge. First, reverse knowledge distillation methods consistently gain ranks under shift, though the benefit depends on the specific distillation design. Second, the sharpest degradation occurs among top-ranked models in normalizing flow, diffusion, and one-class paradigms, suggesting that benchmark overfitting rather than architectural choice is the primary cause. Third, the divergence between absolute and pairwise rankings shows that the two metrics measure different aspects of generalization, limiting the usefulness of single-metric evaluation. DS-MAD should be viewed as complementary to, rather than a replacement for, fixed-benchmark evaluation. Future directions include integrating generalization ability into standard IAD evaluation protocols and extending DS-MAD to other visual tasks.

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